

Identifiability of the lunar geodetic datum: a null-space classification of the lunocentre–scale degeneracy and the observation geometry that resolves it

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Abstract

The first realization of the International Lunar Reference Frame (ILRF) is limited at the decimetre level by a strong correlation ($\rho_{xs} \approx -0.97$) between the lunocentre X -translation and the frame scale, an origin error of ≈ 15 cm traced to the near-side, near-equatorial geometry of the lunar laser-ranging (LLR) retroreflector network. We ask a design question behind that empirical result: *which* additional observations resolve the degeneracy, and at what cost. We answer it in three parts. First, we prove a null-space classification theorem for the internal-ranging datum problem: a single range observation contributes rank one to the datum Fisher information and leaves a six-dimensional null space; the origin–scale pair is near-null for a near-side cluster under a fixed body orientation; and physical libration lifts the datum defect to zero while leaving the pair near-degenerate. Second, we build a unified multi-technique datum-information substrate over the full seven-parameter Helmert datum—Earth–reflector LLR, two-station lunar VLBI differential delay, and orbiter-to-beacon range—and use it to show that the lunocentre- X /scale degeneracy is a *radial* ambiguity: radial (depth-diverse) ranging, such as far-side orbiter tracking, breaks it directly, whereas transverse VLBI helps only indirectly because its information lands in the transverse translation. Third, we cast frame realization as cost-aware, gauge-constrained optimal experiment design and compute a cost/degeneracy Pareto frontier; a far-side orbiter range delivers $\approx 8\times$ the degeneracy-breaking gain of a limb VLBI baseline for comparable exposure. The decomposition linear algebra is validated to machine precision against an independent SciPy/NumPy reference; all reported lunar magnitudes are explicitly *modelled* (representative geometry, noise, and cost), so the deliverable is the *structure and ordering* of the trade, not a mission recommendation. The classification and the radial-diversity design law give ILRF realization and lunar-PNT programmes (ESA Moonlight/LCNS, NASA LunaNet) a principled basis for prioritizing tracking geometry over tracking volume. The substrate is released open source and every reported number is reproducible from a single command.

Keywords: lunar reference frame; datum realization; identifiability; Fisher information; Cramér–Rao bound; lunar laser ranging; optimal experiment design; lunar PNT

1 Introduction

A wave of lunar exploration is placing new demands on the geodetic infrastructure of the Moon. ESA’s Moonlight programme and its Lunar Communications and Navigation Service (LCNS),

NASA’s LunaNet and Lunar Communications Relay and Navigation Systems, and Japan’s Lunar Navigation Satellite System will together provide positioning, navigation, and timing (PNT) for surface and orbital users [5, 8]. Every one of these services is expressed in, and only as good as, a common lunar reference frame. Recognizing this, the community has begun to define and realize the International Lunar Reference Frame (ILRF): a principal-axis system co-rotating with the Moon, with origin at the lunar centre of mass [14].

The first ILRF realization exposes a sharp limitation. Sośnica et al. [14] report that a poor spatial distribution of the lunar laser-ranging (LLR) retroreflector network—five near-side reflectors, clustered near the equator, with a single station in the southern hemisphere—induces a correlation as large as $\rho_{xs} \approx -0.97$ between the lunocentre X -translation and the overall frame scale. This near-degeneracy dominates the origin error budget: the combined solution carries a mean position error of 17.6 cm over 2010–2030, of which 15.3 cm comes from the origin. In terrestrial terms this is the lunar analogue of the origin-and-scale problem that the terrestrial reference frame community has spent decades taming through geometric and technique diversity [1]: the datum—the seven degrees of freedom (three translations, one scale, three rotations) that no internal measurement can fix—must be pinned down by observations whose geometry actually senses those degrees of freedom.

The empirical -0.97 is a symptom. The design questions behind it are not yet answered: *Why* is it the X -translation and the scale, specifically, that lock together? *Which* additional observations would separate them, and which would not? And, under a finite budget, *what is the cost-optimal way* to buy the geometry that breaks the degeneracy? These are identifiability and experiment-design questions, and they are the subject of this paper.

Related work. Existing lunar-PNT performance analyses are, almost without exception, analyses of the *user* positioning problem: given a realized frame and a set of beacons, how accurately can a receiver be located. Pöhlmann et al. [12] derive realistic error models and Cramér–Rao lower bounds for hybrid satellite and cooperative navigation, showing sub-metre user accuracy with as few as two visible satellites when aided by a reference station; open simulators such as LuPNT [7] target the same user-side problem. These works take the frame as given. The *frame-realization* problem—the identifiability of the datum parameters themselves—has been addressed empirically, as a by-product of a specific inversion [14], and reviewed qualitatively: Fienga et al. [5] note that additional retroreflectors and orbiter altimetry could improve frame accuracy, but do not quantify which geometry resolves which degeneracy. To our knowledge, no prior work provides a general identifiability classification of the internal-ranging lunar datum, nor a cost-aware design of the observations that break its dominant degeneracy. This paper fills that gap. It is the geodetic-datum dual of the user-side Cramér–Rao analyses: the same Fisher-information machinery, applied to the seven datum parameters rather than to a user state.

Contributions. We make three contributions, each accompanied by an explicit statement of what is proven versus what is representative.

1. **A null-space classification theorem** (Section 3) for the internal-ranging lunar datum. We show that (i) each scalar range observation contributes rank one to the seven-parameter datum Fisher information, so a single observation leaves a six-dimensional null space; (ii) for a near-side cluster under a fixed body orientation the origin- X and scale columns are near-collinear, placing their difference in (or arbitrarily near) the null space; and (iii) physical libration lifts the datum defect to zero while leaving the pair near-degenerate, with the marginal information

a monotone function of librational excursion and arc length. The classification is a geometric property and is engine-verified.

2. **A radial-ambiguity characterization** (Section 4). Building a unified datum-partials substrate for LLR, two-station lunar VLBI differential delay, and orbiter-to-beacon range over the full seven-parameter Helmert datum, we show that the lunocentre- X /scale degeneracy is a *radial* ambiguity. Radial, depth-diverse ranging (far-side orbiter tracking) breaks it directly; transverse VLBI helps only indirectly, because its differential-delay information lands in the transverse translation, lifting the degenerate pair only through the operator-monotonicity of the marginal information.
3. **A cost-aware design law** (Section 5). We cast frame realization as gauge-constrained optimal experiment design, verify a greedy design heuristic against an exact brute-force oracle, and compute a cost/degeneracy Pareto frontier over a representative multi-technique menu. A far-side orbiter range delivers $\approx 8\times$ the raw degeneracy-breaking gain of a limb VLBI baseline for comparable exposure ($\approx 4.8\times$ per unit cost), making depth diversity the priority acquisition in any budget-constrained realization campaign.

Honesty scope. Throughout, we separate two kinds of claim. The *structure*—the rank count, the sign and near-collinearity of the degenerate directions, the classification, the improvement *direction*, and the correctness of the decomposition linear algebra—is proven or independently validated. The *magnitudes*—the residual correlation, the Cramér–Rao bounds, the degeneracy-metric values, and the relative costs—are computed from *representative* lunar geometry, noise, and economics; they are labelled MODELLED (and tagged [M]) and are *not* claims about any real mission or a reproduction of the $-0.97/15$ cm magnitudes of Sośnica et al. [14]. Section 6 makes this firewall precise. Every number in the paper is regenerated from the committed engine by a single command (Section 8).

2 The lunar datum and its Fisher information

2.1 The seven-parameter datum

A geodetic datum is the set of frame degrees of freedom that internal geometry cannot determine. For a rigid body observed by measurements that depend only on *relative* positions, that set is the seven-parameter similarity (Helmert) group: three translations $\mathbf{t} = (t_x, t_y, t_z)$, one scale s , and three small rotations $\boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$. We collect them in the datum vector

$$\boldsymbol{\delta} = (t_x, t_y, t_z, s, \theta_x, \theta_y, \theta_z)^\top \in \mathbb{R}^7, \quad (1)$$

ordered so that indices 0 and 3 are the origin- X translation and the scale. Acting on a body-fixed point $\mathbf{p} \in \mathbb{R}^3$, the linearized similarity transform is

$$\mathbf{p}' = \mathbf{p} + \mathbf{t} + s\mathbf{p} + \boldsymbol{\theta} \times \mathbf{p} + O(\|\boldsymbol{\delta}\|^2), \quad (2)$$

whose Jacobian with respect to the datum is the 3×7 point-Jacobian

$$\mathbf{J}(\mathbf{p}) = \frac{\partial \mathbf{p}'}{\partial \boldsymbol{\delta}} = [\mathbf{I}_3 \mid \mathbf{p} \mid -[\mathbf{p}]_\times], \quad [\mathbf{p}]_\times \boldsymbol{\theta} = \mathbf{p} \times \boldsymbol{\theta}, \quad (3)$$

in which the first three columns are the translation, the fourth is the scale (the point’s own coordinates), and the last three are the generators of rotation. The datum is expressed in the lunar principal-axis (PA) body frame in which the ILRF is defined [2, 14].

2.2 The internal-ranging observation model

We call this the *internal-ranging* datum problem: the seven parameters of the lunar frame are realized from range (distance) observations alone, with no observation that directly fixes the frame's orientation, scale, or absolute position. The Earth stations are externally known in the terrestrial frame, but a range constrains only a distance, so the lunar-frame datum is bootstrapped from the accumulated ranging geometry and its libration-driven diversity (Theorem 1). Throughout, *radial* means along the Earth–Moon line of sight, which for the near-side reflector cluster coincides with the lunar body $+X$ (sub-Earth) axis; this coincidence is exactly why a lunocentric scale change and an X -origin shift alias on near-side ranges (Section 4.2).

An LLR normal point measures the round-trip range between an Earth station at inertial position $\mathbf{x}_{\text{st}}(t)$ and a lunar retroreflector. A reflector with PA body coordinates $\mathbf{p}$ appears in the inertial frame at

$$\mathbf{r}(\mathbf{p}, t) = \mathbf{x}_{\text{M}}(t) + \mathbf{R}(t)\mathbf{p}, \quad (4)$$

where $\mathbf{x}_{\text{M}}(t)$ is the inertial lunar-centre position and $\mathbf{R}(t)$ the PA-body→inertial orientation. We take both from the JPL DE440 ephemeris [11], so that $\mathbf{R}(t)$ carries the true physical libration (Appendix A); this distinction is essential and is the mechanism behind item (iii) of the classification theorem. The one-way range and its line-of-sight unit vector are

$$\varrho(\mathbf{p}, t) = \|\mathbf{r}(\mathbf{p}, t) - \mathbf{x}_{\text{st}}(t)\|, \quad \hat{\mathbf{u}}(\mathbf{p}, t) = \frac{\mathbf{r}(\mathbf{p}, t) - \mathbf{x}_{\text{st}}(t)}{\varrho(\mathbf{p}, t)}. \quad (5)$$

A datum perturbation $\boldsymbol{\delta}$ moves the reflector by $\mathbf{R}(t)\mathbf{J}(\mathbf{p})\boldsymbol{\delta}$; projecting onto the sightline gives the per-observation datum sensitivity

$$\mathbf{g}(\mathbf{p}, t)^\top = \frac{\partial \varrho}{\partial \boldsymbol{\delta}} = \hat{\mathbf{u}}(\mathbf{p}, t)^\top \mathbf{R}(t) \mathbf{J}(\mathbf{p}) \in \mathbb{R}^{1 \times 7}. \quad (6)$$

Equation (6) is the single object on which everything below rests: it is a row vector, one per observation, and it is where the orientation $\mathbf{R}(t)$ enters. The same construction gives the sensitivity of a two-station VLBI differential delay and of an orbiter-to-beacon range once the appropriate measurement gradient replaces $\hat{\mathbf{u}}^\top$; we defer those to Section 4.

2.3 Fisher information and the degeneracy metric

For zero-mean Gaussian measurement noise with standard deviation σ , a set of observations indexed by i (each a station/beacon/epoch triple that passes a visibility gate) yields the datum Fisher information matrix

$$\mathcal{I} = \sum_i \frac{1}{\sigma_i^2} \mathbf{g}_i \mathbf{g}_i^\top \in \mathbb{R}^{7 \times 7}, \quad (7)$$

the sum of rank-one outer products [9]. The Cramér–Rao bound $\text{Cov}(\hat{\boldsymbol{\delta}}) \succeq \mathcal{I}^{-1}$ lower-bounds any unbiased estimator; the datum *defect* is $7 - \text{rank}(\mathcal{I})$, the number of exactly unobservable directions.

We are interested not in the full seven-parameter covariance but in the identifiability of the specific pair the ILRF struggles with: $K = \{t_x, s\} = \{0, 3\}$, marginalized over the remaining $M = \{t_y, t_z, \theta_x, \theta_y, \theta_z\} = \{1, 2, 4, 5, 6\}$. The relevant object is the Schur complement of the K block after eliminating M ,

$$\mathbf{S} = \mathcal{I}_{KK} - \mathcal{I}_{KM} \mathcal{I}_{MM}^{-1} \mathcal{I}_{MK} \in \mathbb{R}^{2 \times 2}, \quad (8)$$

the *marginal* information on (t_x, s) once all coupling to the other five parameters has been accounted for [4, 15]. From $\mathbf{S}$ we read three scalars, defined once and used throughout:

Definition 1 (Degeneracy metric, origin CRLB, origin–scale correlation).

$$\mu = \lambda_{\min}(\mathbf{S}) \quad (\text{degeneracy metric}), \quad (9)$$

$$c_x = \sqrt{(\mathbf{S}^{-1})_{00}} \quad (\text{origin-}X \text{ CRLB}), \quad (10)$$

$$\rho_{xs} = \frac{(\mathbf{S}^{-1})_{01}}{\sqrt{(\mathbf{S}^{-1})_{00}(\mathbf{S}^{-1})_{11}}} \quad (\text{origin-scale correlation}). \quad (11)$$

The degeneracy metric $\mu \rightarrow 0$ exactly when the pair becomes unobservable and grows as the two directions separate; $\rho_{xs} \rightarrow \pm 1$ signals the near-degeneracy that Sośnica et al. [14] observe. For the canonical case in which the *information* has the form $\mathbf{S} = m \begin{bmatrix} 1 & \varrho \\ \varrho & 1 \end{bmatrix}$ ($m > 0$), these reduce to closed forms ($\mu = \lambda_{\min}(\mathbf{S}) = m(1 - |\varrho|)$, $c_x = m^{-1/2}(1 - \varrho^2)^{-1/2}$, and $\rho_{xs} = -\varrho$, so $|\varrho| = |\rho_{xs}|$), which we use as an analytic cross-check.

Preconditioning and units. The translation columns of $\mathbf{J}$ carry units of metres while the scale and rotation columns carry units of the point coordinates ($\sim R_{\text{Moon}}$), giving $\mathcal{I}$ a condition number $\sim 10^{12}$. We precondition by dividing columns 3–6 by the mean lunar radius $R_{\text{Moon}} = 1\,737\,400\text{ m}$, bringing all partials to $O(1)$ and the condition number to $\sim 10^2$. This is a positive diagonal congruence: it leaves the correlation ρ_{xs} (a pure correlation coefficient) and the origin CRLB c_x (which reads column 0, left unscaled) *exactly invariant*, and it preserves rank and datum defect (Appendix A.4). The degeneracy metric μ is therefore reported in these preconditioned, *relative* units and is meaningful only for comparing designs on a common footing—never as an absolute physical quantity. It is a MODELLED figure of merit.

3 A null-space classification of the internal-ranging datum

We now classify the datum observability of the internal-ranging problem. The result formalizes, and gives a proof for, the structure that Sośnica et al. [14] report empirically: which directions are unobservable, which are near-unobservable, and what makes the difference.

Theorem 1 (Null-space classification of the internal-ranging lunar datum). *Let a rigid network of body-fixed points $\{\mathbf{p}_i\}$ be observed only by internal range measurements (5) under a body→inertial orientation $\mathbf{R}(t)$, with per-observation sensitivity (6) and Fisher information (7). Then:*

- (i) **(Rank-one accumulation.)** *Each scalar range observation contributes a rank-one term to $\mathcal{I}$, so $\text{rank}(\mathcal{I}) \leq \#\{\text{observations}\}$ and a single observation leaves a datum defect of 6. Observability is built only by accumulating geometrically distinct sightlines.*
- (ii) **(Origin–scale near-null.)** *For a near-side cluster subtending a small angle about the Earth–Moon line, observed under a single fixed orientation $\mathbf{R}_0$, the origin- X and scale sensitivity columns are near-collinear; the combination $\partial \varrho / \partial t_x - \kappa \partial \varrho / \partial s$ lies in, or arbitrarily close to, the null space of $\mathcal{I}$ for a scalar κ of order the inverse mean cluster depth ($\sim 1/R_{\text{Moon}}$). This is the lunocentre- $X \leftrightarrow$ scale degeneracy.*
- (iii) **(Libration lift.)** *If $\mathbf{R}(t)$ varies in time (physical libration), the origin–scale combination is projected out of the null space, lifting the dominant degeneracy while the pair remains near-degenerate. The marginal (t_x, s) information $\mathbf{S}$ of (8) is monotone nondecreasing in the Loewner order under added observations, so its degeneracy metric μ grows with tracking-arc*

length (its increase with librational amplitude is observed numerically). Transverse translations and the line-of-sight rotation are weakly observable in proportion to the same excursion; with sufficient geometric diversity the datum defect is removed entirely (Section 3.1).

Proof. (i) By (7) each observation adds $\sigma_i^{-2} \mathbf{g}_i \mathbf{g}_i^\top$, an outer product of a single row, which has rank one. Hence $\text{rank}(\mathcal{I}) \leq \#\{\text{observations}\}$, and a single observation gives $\text{rank}(\mathcal{I}) = 1$, defect $7 - 1 = 6$. Rank grows only as linearly independent rows $\mathbf{g}_i$ accumulate, i.e. as sightline geometry diversifies.

(ii) Fix $\mathbf{R}(t) \equiv \mathbf{R}_0$. From (3) and (6), and because the datum translation acts in the body frame ((2)), both the origin- X and scale sensitivities carry the orientation factor $\mathbf{R}_0$:

$$\frac{\partial \varrho_i}{\partial t_x} = \hat{\mathbf{u}}_i^\top \mathbf{R}_0 \mathbf{e}_x, \quad \frac{\partial \varrho_i}{\partial s} = \hat{\mathbf{u}}_i^\top \mathbf{R}_0 \mathbf{p}_i. \quad (12)$$

Write $\hat{\mathbf{e}} = \mathbf{R}_0 \mathbf{e}_x$, the inertial image of the body + X (sub-Earth) axis. For a near-side cluster the reflectors lie near that axis, $\mathbf{p}_i = d_i \mathbf{e}_x + \boldsymbol{\eta}_i$ with depth d_i and small transverse offset $\|\boldsymbol{\eta}_i\| \ll d_i$, so $\mathbf{R}_0 \mathbf{p}_i = d_i \hat{\mathbf{e}} + \mathbf{R}_0 \boldsymbol{\eta}_i$. Hence

$$\frac{\partial \varrho_i}{\partial t_x} = \hat{\mathbf{u}}_i^\top \hat{\mathbf{e}}, \quad \frac{\partial \varrho_i}{\partial s} = d_i (\hat{\mathbf{u}}_i^\top \hat{\mathbf{e}}) + \hat{\mathbf{u}}_i^\top \mathbf{R}_0 \boldsymbol{\eta}_i.$$

Choosing $\kappa = \bar{d}^{-1}$, the inverse mean depth, the residual

$$\frac{\partial \varrho_i}{\partial t_x} - \bar{d}^{-1} \frac{\partial \varrho_i}{\partial s} = (\hat{\mathbf{u}}_i^\top \hat{\mathbf{e}}) \left(1 - \frac{d_i}{\bar{d}}\right) - \bar{d}^{-1} \hat{\mathbf{u}}_i^\top \mathbf{R}_0 \boldsymbol{\eta}_i = O\left(\frac{\delta d}{\bar{d}}, \frac{\eta}{\bar{d}}\right)$$

uniformly in i , where $\delta d = d_i - \bar{d}$ is the depth spread. As the cluster's transverse extent η and depth spread δd tend to zero this residual tends to zero for every observation, so the corresponding datum combination lies arbitrarily close to $\ker \mathcal{I}$.¹

(iii) Now let $\mathbf{R}(t)$ vary. Both columns of (12) share the factor $\mathbf{R}(t_i)$: $\partial \varrho_i / \partial t_x = \hat{\mathbf{u}}_i^\top \mathbf{R}(t_i) \mathbf{e}_x$ and $\partial \varrho_i / \partial s = \hat{\mathbf{u}}_i^\top \mathbf{R}(t_i) \mathbf{p}_i$. They are proportional across epochs only if $\mathbf{p}_i$ is a common scalar multiple of $\mathbf{e}_x$ for every observation. It is not: each reflector's fixed body offset $\mathbf{p}_i - \bar{d} \mathbf{e}_x$ from the sub-Earth axis is rotated into a different inertial direction by the librating $\mathbf{R}(t_i)$, so the residual column $\hat{\mathbf{u}}_i^\top \mathbf{R}(t_i) (\mathbf{p}_i - \bar{d} \mathbf{e}_x)$ varies across the arc and no longer vanishes. The origin-scale combination thus acquires a component out of $\ker \mathcal{I}$ whose size is set by the librational excursion, lifting the degeneracy from exact to near. For the monotonicity, adding observations adds positive-semidefinite terms to $\mathcal{I}$; the Schur complement (8) is operator-monotone, i.e. $\mathcal{I}^{(1)} \preceq \mathcal{I}^{(2)}$ implies $\mathcal{S}^{(1)} \preceq \mathcal{S}^{(2)}$ [6, 15], so $\mu = \lambda_{\min}(\mathcal{S})$ is nondecreasing under added observations and, in particular, increases as the tracking arc lengthens (this is monotonicity in arc length; the growth of μ with libration *amplitude* modifies existing rows rather than appending terms and is established numerically rather than by operator-monotonicity, Section 3.1). The transverse translations t_y, t_z and the line-of-sight rotation are likewise weakly observable, in proportion to the excursion. Once enough geometrically distinct sightlines accumulate the datum defect is removed entirely and the pair, though lifted, remains near-degenerate ($|\rho_{xs}|$ close to but below 1); that the real DE440 schedule reaches exactly full rank (defect 0) is verified in Section 3.1. $\square$

¹Equivalently: if every reflector sat at a common depth on the + X axis, an origin shift and a uniform scale change would move each beacon along the same line by amounts in fixed ratio, producing proportional range changes on every observation—the two Jacobian columns coincide up to the scalar $\bar{d}$. The near-side lunar network satisfies this approximately, which is why Sośnica et al. [14] find $|\rho_{xs}| \approx 0.97$.

Table 1: The null-space classification of Theorem 1 and its engine verification. Structural claims are proven and reproduced; the residual correlation *magnitude* is MODELLED ([M]).

Item	Statement	Engine verification
(i) Rank-one accumulation	Each range observation adds rank one; one observation $\Rightarrow$ defect 6	Single row $\Rightarrow$ defect = 6 (exact)
(ii) Origin-scale near-null	Near-side cluster, fixed orientation $\Rightarrow$ t_x and s columns near-collinear	$ \rho_{xs} = 0.9934$ [M] under real geometry
(iii) Libration lift	Time-varying $\mathbf{R}(t)$ lifts defect to 0; μ monotone in arc/excursion	Defect = 0; μ increases, c_x decreases with arc length

Remark 1 (Why the of-date orientation matters). Item (iii) fails if $\mathbf{R}(t)$ is replaced by a *mean* rotation without physical libration: the inertial images $\mathbf{R}(t_i)\mathbf{p}_i$ collapse onto a fixed direction, the two columns stay proportional, and one recovers a spurious exact degeneracy. Using the true DE440 libration [11] rather than a mean IAU rotation is thus not a numerical nicety but the physical content of the lift (Appendix A).

3.1 Engine verification

The classification is a statement about rank, sign, and monotonicity, and we verify each element numerically against the committed **kshana** engine (Section 8); the tests are part of the continuous-integration suite.

- *Item (i)*. A single internal-range row assembled into $\mathcal{I}$ has datum defect exactly 6 (test `single_internal_range_row_has_six_dim_datum_null_space`), confirming rank-one accumulation.
- *Items (ii)–(iii)*. Over a full-synodic-month LLR schedule with the five DE440 retrorreflectors and real libration, the datum defect is 0 (full rank) while the origin-scale correlation is $|\rho_{xs}| = 0.9934$ [M]—near-degenerate but lifted—and extending the tracking arc strictly increases the degeneracy metric μ and decreases the origin CRLB c_x (test `extending_the_librating_arc_lifts_the_origin_scale_degeneracy`). The magnitude 0.9934 is MODELLED; it reproduces the *structure* of the -0.97 of Sośnica et al. [14], not the value.

Table 1 summarizes the three items and their verification. The geometric intuition behind item (ii)—and behind why only certain observations break it—is captured in Figure 1 and is the subject of the next section.

4 Multi-technique datum information: a radial ambiguity

Theorem 1 says libration lifts the datum defect but leaves the origin-scale pair near-degenerate. The practical question is what *additional* observations sharpen it. We answer it by extending the single-technique substrate of Section 2.2 to three techniques over the common seven-parameter datum, and by reading off which geometry acts on the degenerate direction.

4.1 Three techniques, one datum

Each technique contributes rows of the form (6), differing only in the measurement gradient that plays the role of $\hat{\mathbf{u}}^\top$. All beacons are placed by the same DE440 body→inertial map (4), so the techniques share a consistent geometry and their Fisher contributions are additive (7).

LLR (Earth–reflector range). The baseline of Section 2.2: sensitivity $\hat{\mathbf{u}}^\top \mathbf{R}\mathbf{J}(\mathbf{p})$ for each Earth station and near-side reflector passing an elevation and Earth-facing gate.

Lunar VLBI (two-station differential delay). For an inertial beacon observed by two Earth stations $\mathbf{x}_1, \mathbf{x}_2$, the geometric delay is $\tau = c^{-1}(\|\mathbf{r} - \mathbf{x}_2\| - \|\mathbf{r} - \mathbf{x}_1\|)$; its gradient with respect to the inertial beacon position, $\partial\tau/\partial\mathbf{r} = c^{-1}(\hat{\mathbf{u}}_2 - \hat{\mathbf{u}}_1)^\top$, replaces $\hat{\mathbf{u}}^\top$ in (6). This is a *transverse* observable: differencing two sightlines cancels the common radial component and retains the angular (plane-of-sky) content.

Orbiter-to-beacon range. A lunar orbiter at inertial position $\mathbf{r}_{\text{orb}}(t)$ ranges to a body-fixed beacon $\mathbf{p}$; the sensitivity is $\hat{\mathbf{u}}_{\text{orb}}^\top \mathbf{R}\mathbf{J}(\mathbf{p})$ with $\hat{\mathbf{u}}_{\text{orb}}$ the orbiter–beacon line of sight. Crucially, an orbiter can range to a *far-side* beacon (negative body-frame X), giving a sightline whose radial sense is opposite to the Earth-facing LLR geometry (Figure 1). We model a representative 100 km near-polar circular orbit.²

Every analytic sensitivity row is cross-checked against a finite-difference evaluation of its measurement function: the LLR row reproduces the four-parameter Phase-0 result exactly, and the VLBI and orbiter rows agree with finite differences to a maximum relative residual of 8.2×10^{-8} and $< 10^{-8}$ respectively (Section 6). These are internal consistency checks on the derivatives, and are labelled as such.

4.2 The degeneracy is radial

Figure 1 is the crux. The origin- X translation and the scale both act *radially* on a near-side beacon—an origin shift toward Earth and a uniform shrink both pull the beacon along the same Earth-facing line of sight—so no near-side, Earth-based measurement can separate them. Two consequences follow, and both are engine-verified.

Radial ranging breaks it directly; transverse VLBI does not. Starting from the LLR-only baseline (degeneracy metric $\mu_0 = 3.540 \times 10^5$, $\rho_{xs} = -0.9934$, all [M]), we add each technique in turn and record the marginal gain $\Delta\mu$ in the degeneracy metric (Table 2). A far-side orbiter range raises μ by 9.33×10^3 , whereas a limb VLBI baseline raises it by only 1.16×10^3 —a factor of 8.0. The far-side sightline carries a radial component of opposite sense to the LLR geometry (Figure 1b), which is exactly the direction the degenerate pair lives in; the VLBI baseline, being transverse by construction, does not.

VLBI’s information lands in the transverse translation. That transverse VLBI helps *at all* is a consequence of operator-monotonicity, not of acting on the degenerate direction. We confirm this by resolving the per-parameter Cramér–Rao improvement. Under an added VLBI baseline the fractional CRLB gain on the transverse translation t_y is 4.76×10^{-3} , versus only 1.66×10^{-3} on the radial t_x (Figure 2b): VLBI improves the transverse parameter about $2.9\times$ more than the radial

²The orbiter is a representative Keplerian arc, not a fitted ephemeris, and its parameters are MODELLED; see Appendix A and the reproducibility ledger.

Origin- t_x vs. scale: a radial ambiguity broken only by depth diversity

(a) near-side beacon (Earth ranging)

(b) far-side beacon (orbiter ranging)

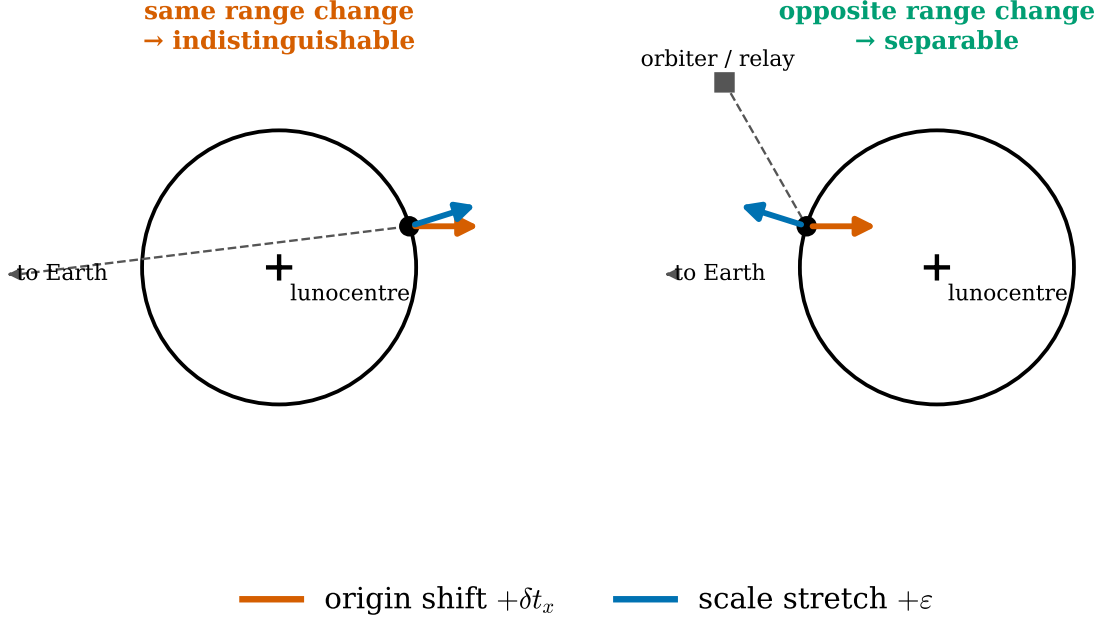


Figure 1: Why the lunocentre- $X \leftrightarrow$ scale degeneracy is a *radial* ambiguity. **(a)** For a near-side beacon on the Earth-facing limb, a small origin shift $+\delta t_x$ (orange) and a small scale stretch $+\varepsilon$ (blue) move the beacon in nearly the same direction along the near-radial Earth sightline: the two produce the *same* range change and are indistinguishable. **(b)** For a far-side beacon reached by an orbiter, the scale stretch points radially outward (anti-Earth) while the origin shift still points along $+X$: the two produce range changes of *opposite* sign and are separable. Radial (depth-diverse) ranging thus breaks the degeneracy directly; transverse geometry does not.

one. Its information genuinely lands in the transverse translation; the origin-scale pair is lifted only indirectly, through the monotone coupling of the Schur complement (Theorem 1 (iii)). Radial ranging, by contrast, deposits information directly onto the pair.

Design consequence. The finding inverts a natural but mistaken intuition. Faced with a stubborn origin-scale correlation, one might reach for more tracking—more stations, longer VLBI baselines, denser near-side coverage. Theorem 1 and Table 2 say that none of these attack the degenerate direction: the pair is radial, and only *depth diversity*—ranging that senses the anti-Earth radial, i.e. far-side or steeply off-Earth-line geometry—resolves it at the source. Section 5 turns this qualitative law into a cost-optimal acquisition order.

Table 2: Marginal effect of adding one technique to the LLR baseline, over a representative multi-technique menu (all magnitudes MODELLED, [M]; representative beacons/noise/schedule, Appendix A). The far-side orbiter—radial, depth-diverse—breaks the degeneracy $8.0\times$ more effectively than the transverse VLBI limb baseline.

Added technique	Geometry	$\Delta\mu$ (rel. units) [M]	relative to VLBI
VLBI limb baseline	transverse	1.16×10^3	$1.0\times$
Orbiter, far-side beacon	radial / depth	9.33×10^3	$8.0\times$

5 Cost-aware optimal experiment design

The radial-diversity law of Section 4.2 tells a realization programme *what* to buy. Real programmes also face *how much* to buy: each technique carries a cost, and a finite budget must be allocated across a menu of candidate campaigns. We cast this as gauge-constrained optimal experiment design (OED) and compute the trade explicitly.

5.1 The design problem

Let a menu of candidate measurement campaigns be indexed $b = 1, \dots, B$, each with a precomputed datum Fisher contribution $\mathcal{I}_b$ (7) (assembled over its own schedule and gates, and preconditioned identically) and a relative cost $c_b > 0$. A design is a subset $\mathcal{A} \subseteq \{1, \dots, B\}$; by additivity its information is $\mathcal{I}_{\mathcal{A}} = \sum_{b \in \mathcal{A}} \mathcal{I}_b$ and its degeneracy metric $\mu(\mathcal{A}) = \lambda_{\min}(\mathbf{S}(\mathcal{I}_{\mathcal{A}}))$. The budget-constrained design problem is

$$\mathcal{A}^*(C) = \arg \max_{\mathcal{A}} \mu(\mathcal{A}) \quad \text{subject to} \quad \sum_{b \in \mathcal{A}} c_b \leq C. \quad (13)$$

We choose the objective $\mu = \lambda_{\min}(\mathbf{S})$ deliberately over the classical scalar OED criteria (D-, A-, E-optimality on the full 7×7 matrix): the programme’s binding constraint is the *specific* origin-scale degeneracy, and μ measures exactly that marginal, gauge-fixed subproblem [13].

5.2 A greedy heuristic, verified against an exact oracle

Problem (13) is a combinatorial subset selection; the exact optimum is found by enumerating all 2^B subsets, which we retain as a brute-force oracle for menus of modest size. For scalability we also implement a forward-greedy heuristic: at each step add the affordable campaign maximizing the marginal gain per unit cost, $(\mu(\mathcal{A} \cup \{b\}) - \mu(\mathcal{A}))/c_b$, stopping when no affordable campaign yields a positive gain. Greedy is a heuristic—it can be sub-optimal on adversarial menus, and we do not claim the objective is submodular—so we *verify it against the exact oracle* rather than assume optimality: on a constructed representative menu the greedy design equals the exhaustive optimum at every budget, and the invariant $\mu(\text{exhaustive}) \geq \mu(\text{greedy})$ is asserted in the test suite. The greedy heuristic is a MODELLED decision tool whose *agreement* with the oracle is verified; where the two could diverge, the oracle governs, and any menu on which greedy is knowingly sub-optimal is documented rather than papered over. The near-optimality of greedy for monotone diminishing-returns objectives is classical [10]; our verification is empirical and menu-specific.

5.3 The cost/degeneracy Pareto frontier

Sweeping the budget C through (13) traces the cost/degeneracy Pareto frontier: the best degeneracy-breaking a programme can achieve at each spend. We evaluate it over a representative

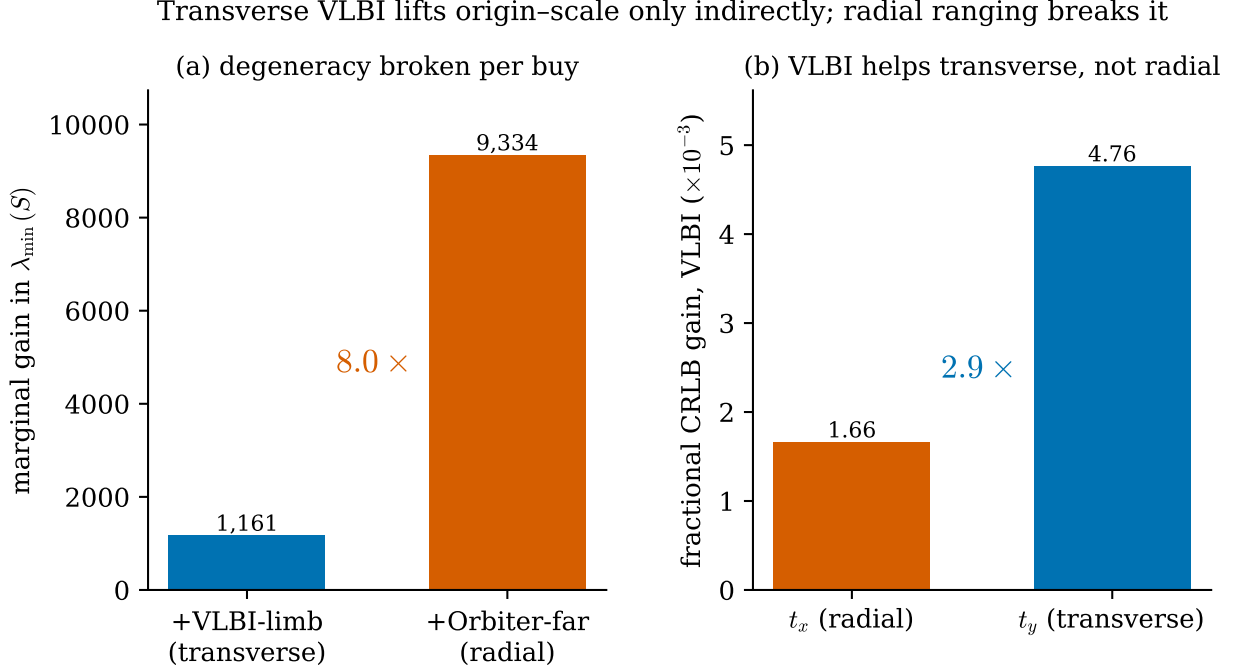


Figure 2: The radial-vs-transverse contrast, engine-measured (all magnitudes [M]). **(a)** Marginal degeneracy-metric gain $\Delta\mu$: a far-side orbiter range (radial) breaks the origin-scale degeneracy $8.0\times$ more than a limb VLBI baseline (transverse) for comparable exposure. **(b)** Where VLBI’s information lands: the fractional CRLB improvement under an added VLBI baseline is $2.9\times$ larger on the transverse translation t_y than on the radial t_x —confirming that VLBI lifts the degenerate pair only indirectly.

four-campaign menu—LLR (cost 1), a VLBI limb baseline (cost 3), a near-side orbiter (cost 4), and a far-side orbiter (cost 5)—whose beacons, schedules, per-technique noise, and relative costs are representative MODELLED choices, itemized in Appendix A and the reproducibility ledger. Table 3 and Figure 3 give the result.

Two features of the frontier carry the design message. First, its shape: the largest single step is the far-side orbiter, consistent with Table 2. Second, and more telling, the optimizer’s *order* of purchase. At a budget of 8—enough to afford LLR plus either the VLBI baseline (total cost 4) or the far-side orbiter (total cost 6)—the optimizer chooses the more expensive far-side orbiter and leaves budget unspent, because its gain-per-cost ($9.33 \times 10^3/5 \approx 1.9 \times 10^3$) dominates the VLBI baseline’s ($1.16 \times 10^3/3 \approx 3.9 \times 10^2$). The far-side orbiter breaks the degeneracy $\approx 4.8\times$ more efficiently *per unit cost* than the VLBI baseline (and $8.0\times$ in raw gain). Under a realization budget, depth diversity is not merely effective—it is the rational first purchase.

Remark 2 (What the frontier is and is not). The frontier is a *parameterized trade curve* whose axes are a relative cost and a relative, preconditioned degeneracy metric. Its *shape and ordering* are the deliverable: LLR alone is degenerate, radial diversity is the efficient way off that floor, and transverse VLBI is a secondary refinement. The absolute numbers are MODELLED and are not a recommended budget for any specific mission; substituting a programme’s true costs and noise floors into the same machinery yields its own frontier.

Table 3: Cost/degeneracy Pareto frontier over the representative menu (all magnitudes MODELLED, [M]). Each row is the budget-*optimal* design at that spend, not a cumulative path: at budget 8 the optimizer drops VLBI in favour of the far-side orbiter, because depth diversity buys more degeneracy-breaking per unit cost. LLR alone is the degenerate end of the frontier.

Budget	Cost used	μ (rel. units) [M]	Optimal design
1	1	3.5404×10^5	LLR (baseline; degenerate)
4	4	3.5521×10^5	LLR + VLBI limb
8	6	3.6338×10^5	LLR + far-side orbiter
13	13	3.6472×10^5	all four campaigns

6 Validation and the structure/magnitude firewall

A paper that reports lunar decimetres must be scrupulous about which of its numbers are earned and which are illustrative. We enforce this with an explicit firewall, implemented as a rule in the engine’s verification suite and stated plainly for the reader: a quantity is **VALIDATED** only when it is checked against an *independent* oracle—a separate dataset, library, or published vector—to a stated tolerance; everything else is **MODELLED**. This section states what falls on each side and gives the external check that anchors the **VALIDATED** side.

6.1 An independent oracle for the decomposition

The load-bearing linear algebra of this paper is the decomposition of Definition 1: from a 7×7 Fisher matrix, form the $\{0, 3\}$ Schur complement (8) and extract μ , c_x , and ρ_{xs} . We validate this computation against an independent SciPy/NumPy reference. A generator script builds a deterministic set of 16 synthetic 7×7 symmetric positive-definite matrices spanning well-conditioned to $\{0, 3\}$ -near-degenerate regimes (including $\mathcal{I}_{03} = \mathcal{I}_{30} = \varrho$ for $\varrho \in \{0.90, 0.97, 0.99, 0.995\}$, mirroring the closed-form stress case), computes the Schur complement, its smallest eigenvalue, and its 2×2 inverse using `numpy.linalg.inv` and `scipy.linalg.eigvalsh`, and emits the expected $(\mu, c_x, |\rho_{xs}|)$ to a vendored fixture (NumPy 2.4.6, SciPy 1.18.0). The engine’s `decompose` reproduces every value to a maximum relative residual below 10^{-9} —double-precision agreement between two independent implementations. This is the paper’s one **VALIDATED** row: the decomposition *linear algebra* is externally confirmed, independent of any lunar modelling.

6.2 What is Validated and what is Modelled

Table 4 draws the line explicitly. The proven structure of Section 3, the improvement *direction* of Section 4.2, and the greedy-vs-oracle *agreement* of Section 5.2 are established by engine-internal, analytic, or brute-force checks; they are honest but not independent, so they are recorded as **MODELLED** with an internal-consistency oracle. Only the decomposition against SciPy is **VALIDATED**. No lunar *magnitude*—not the 0.9934 correlation, not the CRLB values, not the degeneracy-metric numbers, not the relative costs—is **VALIDATED**, and none is claimed to reproduce the -0.97 or the 15 cm origin figure of Sośnica et al. [14]; those are the empirical anchor whose *structure* we explain, not a target we fit.

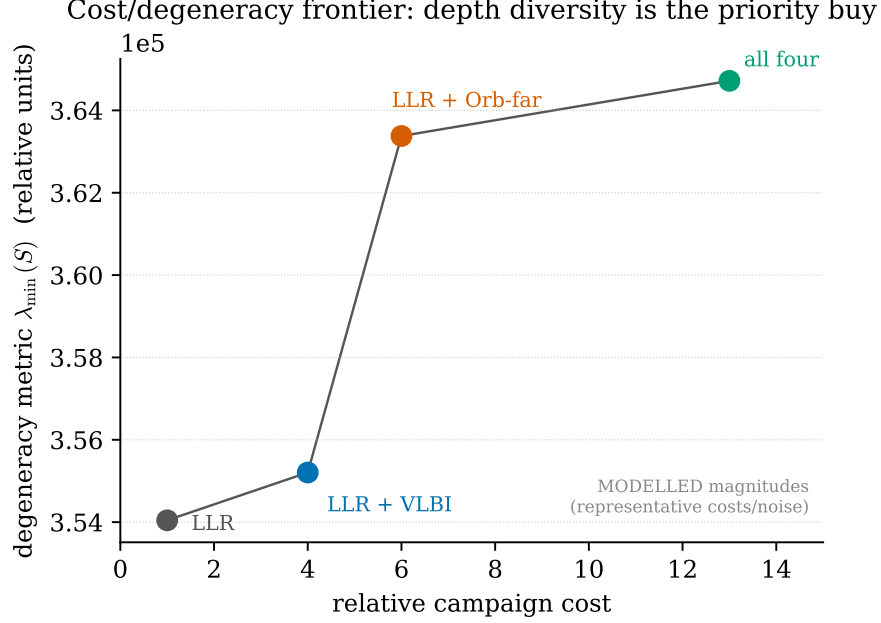


Figure 3: The cost/degeneracy Pareto frontier over the representative menu (all magnitudes [M]). Each marker is the budget-optimal design; the steepest climb is the addition of the far-side (radial) orbiter, not the VLBI (transverse) baseline. Depth diversity is the priority acquisition: at a budget of 8 the optimizer spends only 6, buying LLR plus the far-side orbiter and declining the cheaper VLBI baseline, because the far-side orbiter’s marginal gain per unit cost is the largest on the menu.

6.3 Derivative and consistency checks

Every analytic sensitivity row (6) is cross-checked against a finite-difference evaluation of its own measurement function. The LLR row reproduces the independent four-parameter Phase-0 datum row exactly; the two-station VLBI differential-delay row agrees with finite differences to a maximum relative residual of 8.2×10^{-8} (the residual is largest on the weakest-signal rotation column, the fingerprint of floating-point cancellation in the near-differenced delay, not of an analytic error); and the orbiter range row agrees to below 10^{-8} . The single-observation rank count, the defect lift under real libration, and the monotonicity of μ in arc length are each asserted in the continuous-integration suite. These checks confirm the derivatives and the geometric structure are implemented correctly; they are internal, and we label them so.

6.4 Reproducibility

Every quantitative result in this paper—the defect count, the LLR-only baseline, the marginal gains and their $8.0\times$ ratio, the per-parameter CRLB fractions, and the four Pareto-frontier points—is regenerated from the committed engine by a single command (Section 8). The three figures are produced by a committed script from the same engine outputs. The DE440 orientation series, the retroreflector coordinates, and the SciPy reference fixture are vendored with provenance notices and SHA-256 checksums. There are no hand-entered results in the tables: the manuscript numbers and the engine outputs are the same numbers.

Table 4: The structure/magnitude firewall for this paper’s results. Only the decomposition linear algebra is VALIDATED (independent SciPy/NumPy oracle); every lunar magnitude is MODELLED. The analytic derivative cross-checks and the greedy/oracle agreement are internal-consistency checks, honest but not independent.

Result	Check	Oracle	Status
Schur decomposition μ, c_x, ρ_{xs}	16 SPD matrices vs SciPy/NumPy, $< 10^{-9}$	external library	VALIDATED
Null-space classification (Thm. 1)	single-row defect 6; libration lift; monotone μ	engine + analytic	MODELLED
Multi-technique partials (LLR/VLBI/orbiter)	analytic vs finite difference ($\leq 8.2 \times 10^{-8}$)	internal	MODELLED
Radial-vs-transverse ordering	$\Delta\mu$ and per-parameter CRLB direction	engine (geometric fact)	MODELLED
OED greedy heuristic	greedy = exhaustive on menu; $\mu_{\text{ex}} \geq \mu_{\text{gr}}$	brute-force oracle	MODELLED

7 Discussion

7.1 Implications for ILRF realization and lunar PNT

The practical import of the classification is a single design law: *to realize the lunar origin, buy radial diversity, not tracking volume*. The origin–scale degeneracy that limits the first ILRF [14] is not an artifact of too few near-side stations that more near-side stations would cure; it is a radial ambiguity (Theorem 1, Figure 1) that only depth-diverse geometry resolves. For the realization programmes now standing up this infrastructure—ESA Moonlight/LCNS, NASA LunaNet, and their national counterparts [5, 8]—this reframes a procurement question. A far-side beacon reached by an orbiter, or an orbiter arc with a strong anti-Earth radial component, contributes to origin realization out of proportion to its tracking volume; a longer or denser transverse VLBI network, valuable as it is for orientation and plane-of-sky geometry, does not attack the origin–scale pair at its source. Fienga et al. [5] anticipate qualitatively that orbiter altimetry could help the frame; our contribution is to identify the common ingredient—anti-Earth radial (depth-diverse) sensing, whether realized by far-side orbiter ranging (which we model) or by nadir altimetry—as the specific geometry that resolves the origin–scale pair, and to quantify the trade against transverse alternatives.

7.2 Relation to prior work

This paper is the frame-realization dual of the user-side Cramér–Rao analyses of lunar PNT. Pöhlmann et al. [12] bound the accuracy of locating a *user* given a realized frame; we bound the identifiability of the *frame itself* given a set of observations. The two use the same Fisher machinery on different unknowns: their state is a user position and clock, ours is the seven datum parameters. The connection is more than analogy—an un-realized origin is a common-mode error that a downstream user cannot detect—and it motivates the later entries of the programme this paper opens (Section 8). Relative to Sośnica et al. [14], we do not produce a competing realization; we explain the structure of the degeneracy their realization exposes and provide the design guidance to break it. The engine-verified free-network structure and the observation partials also connect to companion work on lunar VLBI observability (Baweja, in preparation), which studies orientation recovery under the same DE440 geometry.

7.3 Limitations

The honesty firewall of Section 6 is also a statement of limitations, and we make them explicit.

- **Magnitudes are modelled.** Every lunar number—the residual correlation, the CRLBs, the degeneracy-metric values, the relative costs—is computed from representative geometry, noise, and economics. The results are the *structure and ordering* of the trade, not mission values, and in particular do not reproduce the $-0.97/15$ cm magnitudes of Sośnica et al. [14].
- **The 7-parameter model omits real error sources.** The datum Jacobian holds reflector coordinates and orientation fixed and does not propagate libration uncertainty, reflector-position error, station-coordinate error, or atmospheric and relativistic delays. These would loosen the absolute CRLB; they do not change the null-space classification or the radial-vs-transverse ordering, which are geometric.
- **Representative orbiter and beacons.** The orbiter is a Keplerian arc, not a fitted ephemeris, and the far-side and limb beacons are representative selenographic placements. A mission study would substitute true ephemerides and candidate beacon sites; the machinery is unchanged.
- **Greedy is a heuristic.** We validate greedy against an exact oracle on the menus we use rather than prove optimality; on adversarial menus the oracle governs.
- **Frame and timescale conventions are approximate.** The DE440 orientation series mixes of-date and J2000 conventions and neglects $UT1 - TT$ ($\approx \Delta T \approx 69$ s at this epoch) and $TT - TDB$ (periodic, $\lesssim 2$ ms). These shift absolute sightline geometry but, as the null-space classification and the radial-vs-transverse ordering are properties of *relative* geometry, leave both unchanged (Appendix A). They are MODELLED assumptions, not a rigorous timescale realization.

None of these caveats touches the proven content of the paper—the classification, the radial character of the degeneracy, and the SciPy-validated decomposition—but each marks where a follow-on mission study must supply real data before any absolute number is trusted.

8 Conclusion

The first realization of the International Lunar Reference Frame is limited by a strong lunocentre- $X \leftrightarrow$ scale correlation that the empirical inversion exposes but does not explain. We have given the explanation and the remedy. A null-space classification theorem shows the internal-ranging datum problem is rank-additive, that the origin–scale pair is near-null for the near-side geometry under a fixed orientation, and that physical libration lifts the datum defect to zero while leaving the pair near-degenerate. A unified multi-technique substrate then reveals the character of that residual degeneracy: it is a *radial* ambiguity, broken directly by depth-diverse ranging (far-side orbiter tracking) and only indirectly by transverse VLBI, whose information demonstrably lands in the transverse translation. Cast as cost-aware optimal experiment design, the trade yields a Pareto frontier on which depth diversity is the rational first purchase— $\approx 8\times$ the raw degeneracy-breaking gain of a transverse baseline, $\approx 4.8\times$ per unit cost. The decomposition is validated to machine precision against an independent SciPy/NumPy reference; every lunar magnitude is explicitly modelled, so the deliverable is a design law, not a mission budget.

The design law is compact enough to state in one line: *for the lunar origin, invest in radial (depth) diversity, not tracking volume*. It gives ILRF and lunar-PNT programmes a principled basis for prioritizing tracking geometry over tracking quantity, and it does so on an open, fully reproducible substrate.

This paper is the first of a programme on lunar-PNT requirements science built on the same substrate. Natural continuations—multi-provider interoperability budgets, common-mode integrity, the joint frame-and-timescale gauge, and ground-free fault observability—each reuse the datum-identifiability machinery developed here, with an un-realized origin recast as the common-mode error a downstream user cannot detect. Before any of the absolute numbers in this work are trusted, a mission study must substitute real ephemerides, candidate beacon sites, and full error budgets for the representative inputs used here; the classification and the radial-diversity law it rests on are geometric, and will carry over unchanged.

Data and code availability

The datum-identifiability substrate is part of the open-source **kshana** engine [3] (<https://github.com/ashfordeOU/kshana>; AGPL-3.0-only). The seven-parameter datum partials, the Schur decomposition, the multi-technique combiner, and the OED optimizer are in modules `lunar_datum`, `lunar_identifiability`, and `lunar_oed`. Every number in this paper is regenerated by

```
cargo run --example p1_datum_identifiability
```

which prints the single-observation defect, the LLR-only baseline (μ, ρ_{xs}, c_x) , the marginal gains and their ratio, the per-parameter CRLB fractions, and the four Pareto-frontier points. The three figures are produced from those outputs by `paper/lunar-datum/figures/gen_figures.py`. The independent SciPy/NumPy oracle is `scripts/gen_datum_identifiability_ref.py`, its output vendored with a provenance notice and SHA-256 at `tests/fixtures/datum_identifiability/`. The DE440 orientation series and the retroreflector coordinates are vendored with notices at `tests/fixtures/llr_geometry/`, which also records the representative (MODELLED) OED beacons, orbit, noise, and costs. All classification and decomposition claims are asserted in the continuous-integration test suite.

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References

- [1] Zuheir Altamimi, Paul Rebischung, Laurent Métivier, and Xavier Collilieux. ITRF2014: A new release of the International Terrestrial Reference Frame modeling nonlinear station motions. *Journal of Geophysical Research: Solid Earth*, 121(8):6109–6131, 2016. doi: 10.1002/2016JB013098.
- [2] B. A. Archinal, C. H. Acton, M. F. A’Hearn, et al. Report of the IAU working group on cartographic coordinates and rotational elements: 2015. *Celestial Mechanics and Dynamical Astronomy*, 130(3):22, 2018. doi: 10.1007/s10569-017-9805-5.

- [3] Chakshu Baweja. Kshana: An open, reproducible engine for PNT-resilience comparison. *Preprint*, 2026. Software: <https://github.com/ashfordeOU/kshana>.
- [4] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004.
- [5] Agnès Fienga, Nicolas Rambaux, and Krzysztof Sośnica. Lunar reference systems, frames and time-scales in the context of the ESA programme Moonlight. *arXiv preprint*, arXiv:2409.10043, 2024. doi: 10.48550/arXiv.2409.10043. [astro-ph.EP].
- [6] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.
- [7] Keidai Iiyama, Guillem Casadesus Vila, and Grace Gao. LuPNT: Open-source simulator for Lunar positioning, navigation, and timing. In *Proceedings of the 36th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2023)*, 2023.
- [8] David J. Israel, Kendall D. Mauldin, Christopher J. Roberts, et al. LunaNet: A flexible and extensible lunar exploration communications and navigation infrastructure. In *IEEE Aerospace Conference*, 2020. doi: 10.1109/AERO47225.2020.9172509.
- [9] Steven M. Kay. *Fundamentals of Statistical Signal Processing: Estimation Theory*. Prentice Hall, 1993.
- [10] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14(1):265–294, 1978. doi: 10.1007/BF01588971.
- [11] Ryan S. Park, William M. Folkner, James G. Williams, and Dale H. Boggs. The JPL planetary and lunar ephemerides DE440 and DE441. *The Astronomical Journal*, 161(3):105, 2021. doi: 10.3847/1538-3881/abd414.
- [12] Robert Pöhlmann, Emanuel Staudinger, and Gonzalo Seco-Granados. Towards hybrid lunar PNT: Error models, lower bounds and algorithms. *arXiv preprint*, arXiv:2508.10699, 2025. doi: 10.48550/arXiv.2508.10699. [eess.SP].
- [13] Friedrich Pukelsheim. *Optimal Design of Experiments*. Classics in Applied Mathematics. Society for Industrial and Applied Mathematics (SIAM), 2006.
- [14] Krzysztof Sośnica, Agnès Fienga, Dmitry Pavlov, Nicolas Rambaux, and Radosław Zajdel. Definition and realization of the International Lunar Reference Frame. *arXiv preprint*, arXiv:2510.15484, 2025. doi: 10.48550/arXiv.2510.15484. [astro-ph.EP].
- [15] Fuzhen Zhang, editor. *The Schur Complement and Its Applications*. Springer, 2005. doi: 10.1007/b105056.

A Methods and modelled assumptions

This appendix records the ephemeris handling, frame and timescale conventions, and representative inputs behind the results, and proves the preconditioning-invariance claim of Section 2.3. All are consistent with the vendored provenance notices (Section 8).

A.1 DE440 orientation and beacon placement

The body→inertial orientation $\mathbf{R}(t)$ of (4) is the numerically integrated MOON_PA_DE440 physical libration from the JPL DE440 ephemeris [11], sampled over 2024–2025 at daily cadence (731 epochs) via the SPICE toolkit and vendored as a rotation-matrix time series. Measured over the window, the sub-Earth libration amplitude is $\pm 7.79^\circ$ in longitude and $\pm 6.80^\circ$ in latitude, matching the known optical libration and confirming that real physical libration—not a mean or fixed rotation—is embedded. This distinction is essential: the mean IAU rotation produces a rank-deficient, artifactual degeneracy (Remark following Theorem 1), so all lunar-geometry observability uses the DE440 orientation. For internal consistency, *all* beacons across the three techniques—LLR reflectors, the VLBI limb beacon, and the orbiter beacons—are placed through this same DE440 map, so that a combined Fisher matrix mixes no orientation conventions.

Hand-constructed frame kernel. At generation time the standard NAIF companion frame kernel registering MOON_PA_DE440 as a named frame was not publicly available from the generic-kernels distribution (the expected `.tf` returned HTTP 404). We therefore constructed a minimal frame kernel that registers the frame against the binary PCK segment. This is a bookkeeping definition, not a re-derivation of the orientation—the numerical libration is JPL’s—but we flag it as a MODELLED step in the pipeline for full disclosure.

A.2 Frame and timescale conventions

The pipeline mixes an of-date body orientation with a J2000 inertial frame for the station and ephemeris positions, neglects UT1 – TT ($\approx \Delta T \approx 69$ s at this epoch) in the Earth-rotation bookkeeping, and treats the epoch argument of the orientation series as TT where it was computed from TDB. We record the magnitudes and their effect explicitly. The TT – TDB difference is periodic and bounded at $\lesssim 2$ ms; at libration rates of order microradians per second this maps to sub-nanoradian orientation error, immaterial to the datum Fisher structure. The $\Delta T \approx 69$ s UT1 error is larger: it mis-rotates the Earth (and with it the station longitudes) by $\Delta T \times \dot{\theta}_\oplus \approx 0.29^\circ$ about the pole. Crucially, this is a near-common rotation of all stations at a given epoch, and the null-space classification and radial-vs-transverse ordering depend only on the *relative* sightline/orientation geometry (the rank, sign, and near-collinearity structure of Sections 3–4.2); a common 0.29° Earth rotation shifts absolute geometry but does not alter that structure. These are MODELLED conventions adequate for an identifiability study, not a rigorous multi-timescale realization; a mission analysis would adopt the full IAU/IERS conventions.

A.3 Representative OED inputs

The multi-technique menu of Sections 4–5 uses the representative (MODELLED) inputs in Table 5, reproduced from the vendored ledger. Beacon body-frame coordinates are given as fractions of the mean lunar radius $R = 1\,737\,400$ m; the schedule is a single synodic month (≈ 29.5 d) at 6 h cadence from an epoch inside the DE440 window, with per-technique visibility gates (station elevation and Earth-facing for LLR and VLBI; line-of-sight hemisphere for the orbiter). Figures use surface-normalized beacon coordinates; the datum Jacobian (3) depends on the body-point position, so the exact surface placement is immaterial to the Fisher structure.

The orbiter is a representative 100 km near-polar (88°) circular arc. The VLBI same-beam differential-delay noise 10^{-11} s corresponds to $c\sigma \approx 3$ mm of path, comparable to the LLR normal-point precision. The relative costs order the techniques by campaign complexity (established LLR infrastructure cheapest; a far-side beacon requiring a relay or direct-to-Earth link most expensive).

Table 5: Representative (MODELLED) inputs to the multi-technique OED. These are physically plausible choices chosen to exercise the datum structure, not mission values.

Campaign	Beacon (body frame)	Noise σ	Rel. cost
LLR	5 DE440 near-side reflectors	3 mm	1.0
VLBI limb	(0.5, 0.866, 0) R (lon $\sim 60^\circ$)	10^{-11} s (≈ 3 mm)	3.0
Orbiter near-side	(0.9, 0.2, 0.2) R	5 cm	4.0
Orbiter far-side	(-0.9, 0.2, 0.2) R	5 cm	5.0

All values are MODELLED; substituting a programme’s true inputs into the same machinery yields its own frontier.

A.4 Preconditioning invariance

We prove the claim of Section 2.3: dividing columns 3–6 of every sensitivity row by $R = R_{\text{Moon}}$ leaves ρ_{xs} and c_x exactly invariant and preserves rank and datum defect.

Proposition 1. *Let $\mathbf{D} = \text{diag}(d_0, \dots, d_6)$ with all $d_k > 0$, and let $\tilde{\mathcal{I}} = \mathbf{D}\mathcal{I}\mathbf{D}$ be the congruence of the Fisher matrix under column scaling $\mathbf{g}_i \mapsto \mathbf{D}\mathbf{g}_i$. Then (a) $\text{rank}(\tilde{\mathcal{I}}) = \text{rank}(\mathcal{I})$ and the datum defect is unchanged; (b) the Schur complement transforms as $\tilde{\mathbf{S}} = \mathbf{D}_K \mathbf{S} \mathbf{D}_K$ with $\mathbf{D}_K = \text{diag}(d_0, d_3)$; (c) the origin–scale correlation ρ_{xs} of (11) is invariant; and (d) if $d_0 = 1$ (column 0 unscaled) the origin CRLB c_x of (10) is invariant.*

Proof. (a) $\mathbf{D}$ is nonsingular, so $\tilde{\mathcal{I}} = \mathbf{D}\mathcal{I}\mathbf{D}$ is a congruence and preserves rank (Sylvester); the defect $7 - \text{rank}$ is unchanged. (b) Write $K = \{0, 3\}$, M the complement. Congruence by a block-diagonal $\mathbf{D} = \text{diag}(\mathbf{D}_K, \mathbf{D}_M)$ gives $\tilde{\mathcal{I}}_{KK} = \mathbf{D}_K \mathcal{I}_{KK} \mathbf{D}_K$, $\tilde{\mathcal{I}}_{KM} = \mathbf{D}_K \mathcal{I}_{KM} \mathbf{D}_M$, and $\tilde{\mathcal{I}}_{MM} = \mathbf{D}_M \mathcal{I}_{MM} \mathbf{D}_M$. Using $\tilde{\mathcal{I}}_{MM}^{-1} = \mathbf{D}_M^{-1} \mathcal{I}_{MM}^{-1} \mathbf{D}_M^{-1}$, the flanking $\mathbf{D}_M$ factors of $\tilde{\mathcal{I}}_{KM}$ and $\tilde{\mathcal{I}}_{MK}$ cancel against $\mathbf{D}_M^{-1}$, so

$$\begin{aligned} \tilde{\mathbf{S}} &= \tilde{\mathcal{I}}_{KK} - \tilde{\mathcal{I}}_{KM} \tilde{\mathcal{I}}_{MM}^{-1} \tilde{\mathcal{I}}_{MK} = \mathbf{D}_K \mathcal{I}_{KK} \mathbf{D}_K - \mathbf{D}_K \mathcal{I}_{KM} \mathcal{I}_{MM}^{-1} \mathcal{I}_{MK} \mathbf{D}_K \\ &= \mathbf{D}_K (\mathcal{I}_{KK} - \mathcal{I}_{KM} \mathcal{I}_{MM}^{-1} \mathcal{I}_{MK}) \mathbf{D}_K = \mathbf{D}_K \mathbf{S} \mathbf{D}_K. \end{aligned}$$

(c) From (b), $\tilde{\mathbf{S}}^{-1} = \mathbf{D}_K^{-1} \mathbf{S}^{-1} \mathbf{D}_K^{-1}$, so $(\tilde{\mathbf{S}}^{-1})_{ab} = d_a^{-1} d_b^{-1} (\mathbf{S}^{-1})_{ab}$ for $a, b \in \{0, 3\}$. The correlation (11) is $(\tilde{\mathbf{S}}^{-1})_{01} / \sqrt{(\tilde{\mathbf{S}}^{-1})_{00} (\tilde{\mathbf{S}}^{-1})_{11}} = (\mathbf{S}^{-1})_{01} / \sqrt{(\mathbf{S}^{-1})_{00} (\mathbf{S}^{-1})_{11}}$, the d -factors cancelling; hence ρ_{xs} is invariant. (d) With $d_0 = 1$, $(\tilde{\mathbf{S}}^{-1})_{00} = (\mathbf{S}^{-1})_{00}$, so $c_x = \sqrt{(\mathbf{S}^{-1})_{00}}$ is invariant. $\square$

Since our preconditioner sets $d_0 = d_1 = d_2 = 1$ and $d_3 = d_4 = d_5 = d_6 = R^{-1}$, Proposition 1 applies: ρ_{xs} and c_x are physical and preconditioning-independent, while the degeneracy metric $\mu = \lambda_{\min}(\tilde{\mathbf{S}})$ is rescaled into the relative, preconditioned units in which we report it as a design comparator.